

Keith Aumiller, MBA, BS Psychology

Business Data Engineering & AI Leader for Enterprise Transformation

Philadelphia, PA | (314) 369-0811

Email: keith.aumiller@stlouisintegration.com

<https://stlouisintegration.com> | <https://github.com/keithaumiller>

[LinkedIn](#): 14k followers | Administrator: [Artificial Intelligence and Deep Learning](#) 600k+ member

Executive Profile

Data and AI transformation leader with 20+ years architecting enterprise-wide data governance solutions and building scalable data organizations. Deep expertise in pharmaceutical and healthcare environments, having led the data services organization responsible for 700+ clinical trials for top 20 pharmaceutical companies and established regulatory-compliant data quality frameworks. Expert in transforming traditional organizations into data-driven enterprises through comprehensive data infrastructure and AI capabilities. Proven track record collaborating with existing leadership teams to build consensus, secure buy-in, and execute transformational roadmaps generating \$20M+ in new revenue streams.

Organizational Design and Development

I excel at designing and implementing data services organizations that drive measurable business impact. I have a proven ability to architect scalable data infrastructures and build high-performing teams. I transform fragmented data operations into unified, agile organizations. My expertise spans the full lifecycle of data services—from strategic vision and process design to technology selection, governance, and operational rollout—ensuring that every solution is aligned with business objectives and regulatory requirements. My collaborative leadership style fosters consensus and buy-in across stakeholders, enabling successful adoption of new data platforms and analytics capabilities that deliver sustained value.

Strategic Differentiators

- **Healthcare & Pharmaceutical Leadership** - Led data services initiatives for 700+ clinical trials with comprehensive regulatory compliance and quality frameworks
- **Enterprise Data Engineering** - Designed and implemented scalable data infrastructure across multiple industries including healthcare, energy, and financial services
- **Data Quality & Governance** - Established comprehensive data governance frameworks ensuring regulatory compliance and operational excellence
- **AI Vision & Transformation** - Successfully collaborated with existing teams to build new AI and data capabilities without organizational disruption

Professional Experience

St. Louis Integration LLC – Data Science & AI Consulting Practice

June 2007 - Present | Philadelphia, PA

<https://stlouisintegration.com>

Provides specialized executive consulting for data engineering, AI strategy, and organizational transformation. Working with Fortune 50 companies, startups, healthcare systems, and government agencies on mission-critical data initiatives. Clients

included AbbVie, Center for Disease Control, AmeriGas, and multiple startups (Sentiosearch, MymobileHealthcare, qtspecialists)

Signant Health

Director Data Science Services Products

February 2018 - November 2021 | Philadelphia, PA

Led 100+ person data science practice delivering AI/ML solutions for 700+ large-scale clinical trials conducted by top 20 pharmaceutical companies. Built and scaled consulting practice specializing in Life Sciences, Pharma R&D, and Medical Devices.

Pharmaceutical Data Engineering & Clinical Infrastructure:

- Transformed single-analyst data function into comprehensive \$20M+ AI and data services organization serving pharmaceutical industry in one year
- Designed enterprise AI architectures for Fortune 50 pharmaceutical clients, implementing comprehensive data governance principles suitable for clinical research environments
- Built scalable data infrastructure and AI platform architecture supporting 700+ clinical trial operations and analytics workloads
- Established MLOps and data product management practices enabling rapid deployment of AI-powered solutions for pharmaceutical applications

Clinical Research Data Quality & Compliance:

- Developed flagship SMARTSIGNALS platform - comprehensive AI-powered risk management solution for clinical trial oversight and pharmaceutical operations
- Led statistical analysis initiatives ensuring scientific rigor in model development, validation, and deployment processes for pharmaceutical applications
- Implemented comprehensive data governance standards and quality monitoring processes ensuring reliable, compliant data products
- Managed cross-functional teams of PhD-level data scientists and ML engineers, achieving dramatically improved customer satisfaction across pharmaceutical client engagements measured by open issues reduction of 80%

Revenue Growth & Strategic Development:

- Developed go-to-market strategy for AI and data products contributing to securing multiple eight-figure contracts with pharmaceutical companies
- Built revenue models and pricing strategies for data products balancing market competitiveness with sustainable profit margins
- Established customer success frameworks and feedback loops informing product roadmap for pharmaceutical market needs

AbbVie

Clinical Research Organization and Clinical Data Management Vendor Management Consultant

June 2022 - June 2023 | Philadelphia, PA

- One-year contract to establish oncology early development data management pipeline.
- Managed multiple Clinical Research Organizations (CROs) for delivery and timeline efficacy on oncology drug development monitoring programs
- Established quality controls and accountability frameworks ensuring technical delivery excellence for clinical data management
- Implemented comprehensive vendor management processes for pharmaceutical clinical research operations

Center for Disease Control

Data Science and AI Infrastructure Consultant

January 2022 - June 2023 | Philadelphia, PA

- Two-year contract supporting establishment of Office of Information Technology within NIOSH division.
- Collaborated with NIOSH leadership team to design architecture for AI infrastructure supporting national health research initiatives
- Worked with National Personal Protective Technology Laboratory, Health Effects Laboratory, and Division of Field Studies and Engineering
- Successfully delivered National Firefighter Registry project on-time and on-budget after inheriting the behind-schedule initiative

NRG

Data Program and Product Manager

August 2014 - December 2016 | Philadelphia, PA

Data Strategy & Infrastructure Transformation:

- Established enterprise data strategy and infrastructure roadmap positioning NRG for digital transformation and market evolution
- Developed comprehensive business case for data platform investments securing executive approval and multi-year budget commitments
- Led system integration initiatives unifying disparate data sources into coherent enterprise data ecosystem

Product Development & Innovation:

- Developed machine learning and advanced analytics capabilities integral to enterprise profitability analysis and strategic planning
- Built data products and analytical tools enabling business users to access insights and make data-driven decisions independently
- Managed analytics components of M&A due diligence across 10+ acquisitions, providing data-driven insights for strategic decision-making

Citigroup

Vice President Treasury and Trade Solutions

November 2011 - July 2014 | Philadelphia, PA

Enterprise Data Platform & AI Integration:

- Led transformation of traditional financial operations into AI-powered platform managing \$1B+ daily transactions
- Developed comprehensive AI and data strategy for treasury operations improving fraud detection accuracy and operational efficiency
- Built cross-functional organization of 45+ professionals spanning data science, engineering, operations, and business analysis

Data Products & Operational Intelligence:

- Created AI-powered fraud monitoring and risk management products using machine learning models and behavioral analytics
- Developed real-time data products and dashboards providing operational visibility and enabling proactive decision-making
- Built data integration solutions connecting disparate financial systems into unified platform for comprehensive business intelligence

Early Career & Foundation Building

2000-2011 | Multiple Locations

Built foundational expertise in organizational management, program management, statistics systems, enterprise data systems, integration architecture, and large-scale technology transformation across multiple industries including financial services (Edward Jones Investments - 5 years), payment systems (MasterCard), healthcare technology (Express Scripts), aerospace (Boeing), and financial markets (Bridge Information Systems).

Education

Washington University in St. Louis, Olin School of Business

Master of Business Administration (MBA) | August 2009 - May 2011

Truman State University, Kirksville, MO

Bachelor of Science in Psychology | August 1996 - May 2000

Technical Expertise

Data Engineering & Architecture:

Enterprise Data Architecture, Cloud-Native Platforms (AWS, Azure, GCP), Data Lake and Warehouse Design, MLOps and DataOps, API Strategy, Medallion Architecture, ETL/ELT Systems

Healthcare & Pharmaceutical:

Clinical Data Management, Regulatory Compliance Frameworks, Clinical Trial Analytics, Pharmaceutical Data Governance, Statistical Modeling for Life Sciences, Biostatistics

Data Quality & Governance:

Data Governance Frameworks, Data Stewardship, Master Data Management, Data Quality Monitoring, Regulatory Compliance, Audit and Validation Processes

Advanced Analytics & AI:

Machine Learning Strategy, Generative AI Implementation, Statistical Modeling, Predictive Analytics, Deep Learning Frameworks (TensorFlow, PyTorch), MLOps

Technology Platforms:

SQL, Python, R, Databricks, Snowflake, Dataiku, Matillion, Tableau, QLIK, Clinical Research Platforms

Demonstration Project

<https://thetruthperspective.org> - AWS hosted GenAI solution utilizing an open source content management system, generative AI content generation, social media integration, sentiment analysis and more